



UPY 
**UNIVERSIDAD
POLITÉCNICA
DE YUCATÁN**

AI-Powered study game

PROGRAMMING

Nathan Eliel Balam Canto

Data and AI Engineering

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Previous state of the project

- **Core Concept:** A system that extracts text from notes to generate multiple-choice questions.
- **Logic Flaws:** The main loop relied on WHILE game_active IS TRUE without a proper exit condition if the questions ran out.
- **Data Processing:** The AI ingested the entire sourceMaterialText at once, regardless of length.
- **Output Output:** Binary feedback. If a user failed, the system only displayed "Incorrect feedback" and moved on.

Progress and improvements

- **Core Concept:** A system that extracts text from notes to generate multiple-choice questions.
- **Logic Flaws:** The main loop relied on WHILE game_active IS TRUE without a proper exit condition if the questions ran out.
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- **Output Output:** Binary feedback. If a user failed, the system only displayed "Incorrect feedback" and moved on.

Application of feedback (Part 1 - Technical Logic)

- **Peer Suggestion:** "In your WHILE loop, you never check if questions are over... it will run forever and crash with an out-of-bounds error."
- **Action Taken:** Modified the loop condition to check the length of the question queue.
- **Updated Code Logic:**
WHILE index < length(question_queue)

Application of feedback (Part 2 - AI limits & UX)

- **Peer Suggestion:** "Processing a long PDF all at once will cause token limit issues... implement a step of chunking."
- **Action Taken:** Added a pre-processing step before the extraction phase.
- **Updated Code Logic:** `chunked_text <- SPLIT sourceMaterialText INTO manageable_blocks`
- **Peer Suggestion:** "It would be better if the program also prints the correct answer and a brief explanation."
- **Action Taken:** Updated the output interface to retrieve and display the correct answer and its rationale when the user fails.

Application of feedback (Part 3 - Gamification)

- **Peer Suggestion:** "I propose keeping a motivation like streaks... add rewards or achievements."
- **Action Taken:** Added a `current_streak` variable that resets upon a wrong answer and multiplies the final score to increase engagement.
- **Updated Code Logic:**
IF correct THEN `current_streak <- current_streak + 1`

Acknowledgements

Thank you to all my classmates for the rigorous technical review. The suggestions directly prevented memory errors and significantly improved the data pipeline of this project.

Form response summary

- **Data Collected:** 23 detailed peer reviews from UPY Engineering students.
- **Key Finding 1 (Technical):** Multiple students correctly identified critical logical errors in the initial pseudocode (infinite loops, missing line breaks).
- **Key Finding 2 (AI Feasibility):** High concern regarding how the AI handles large inputs and prevents hallucinations, validating the need for the new chunking and strict-context generation features.
- **Key Finding 3 (User Experience):** 100% agreement that gamification makes studying more dynamic, but users heavily demanded detailed explanations for incorrect answers over simple scores.